Retail Sales Forecasting & Analytics Dashboard

Interactive machine learning dashboard for forecasting, model evaluation, HyperOpt optimization, and MLflow experiment tracking.

Best Model

Tuned Ridge Regression

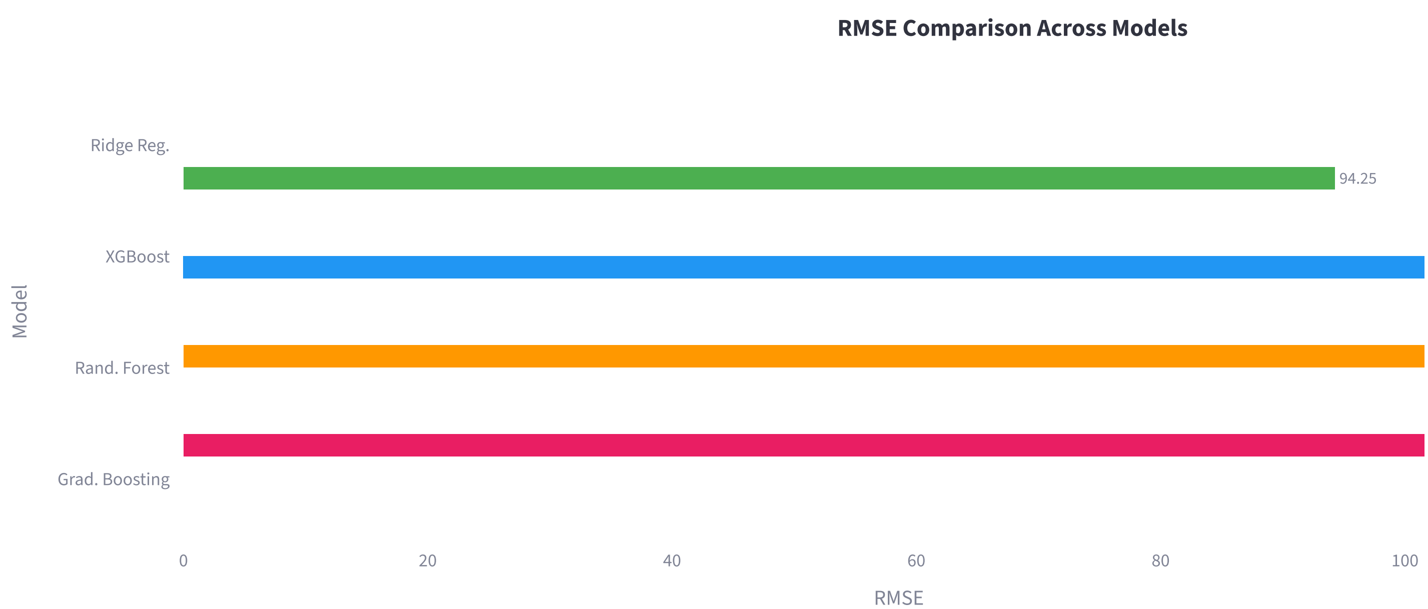
Best Model Metrics

MAE	RMSE	MAPE	R^2
75.93	94.25	19.69%	0.651

Model Comparison

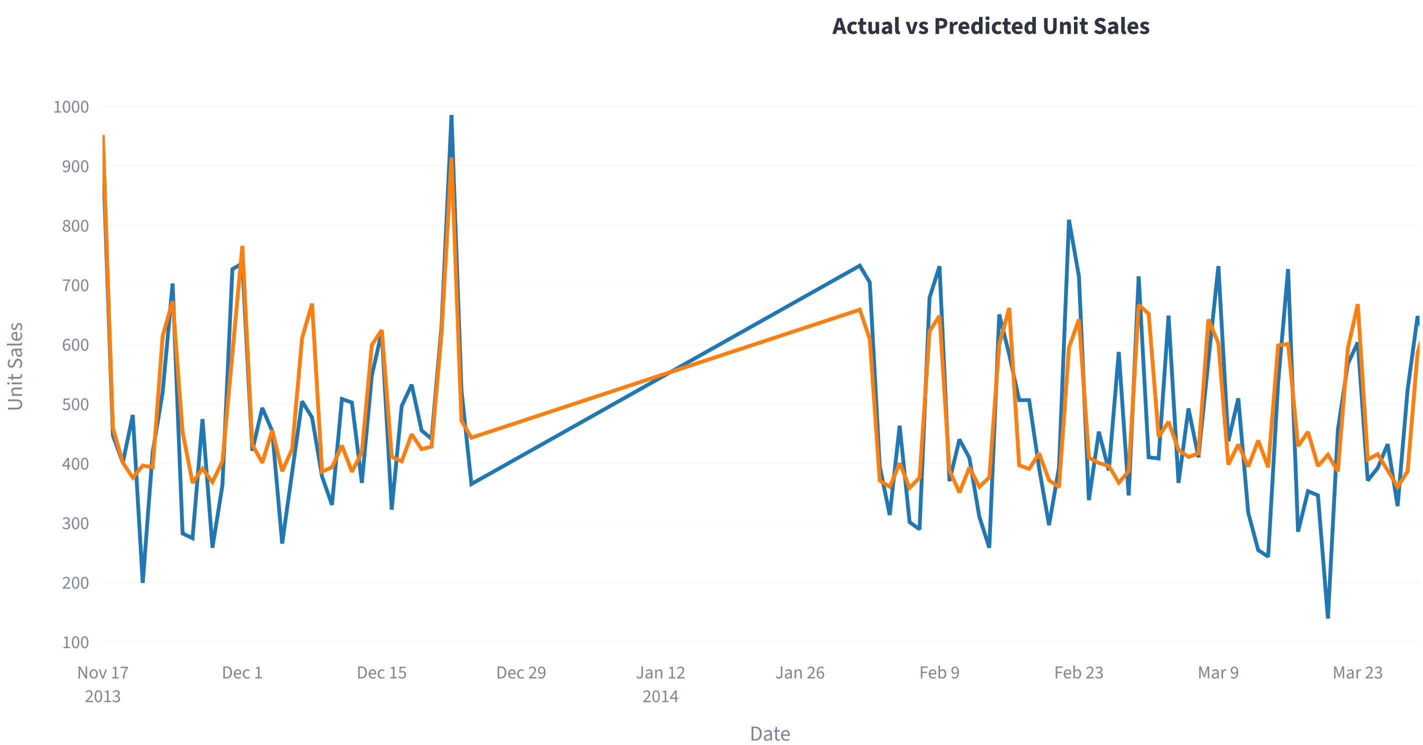
Tuned model performance:

Model	MAE	RMSE	MAPE	R2
Tuned Ridge Regression	75.9349	94.2536	19.6877	
Tuned XGBoost	83.8619	107.5245	20.7241	
Tuned Random Forest	82.2169	107.8448	20.5765	
Tuned Gradient Boosting	85.5627	111.2754	21.168	



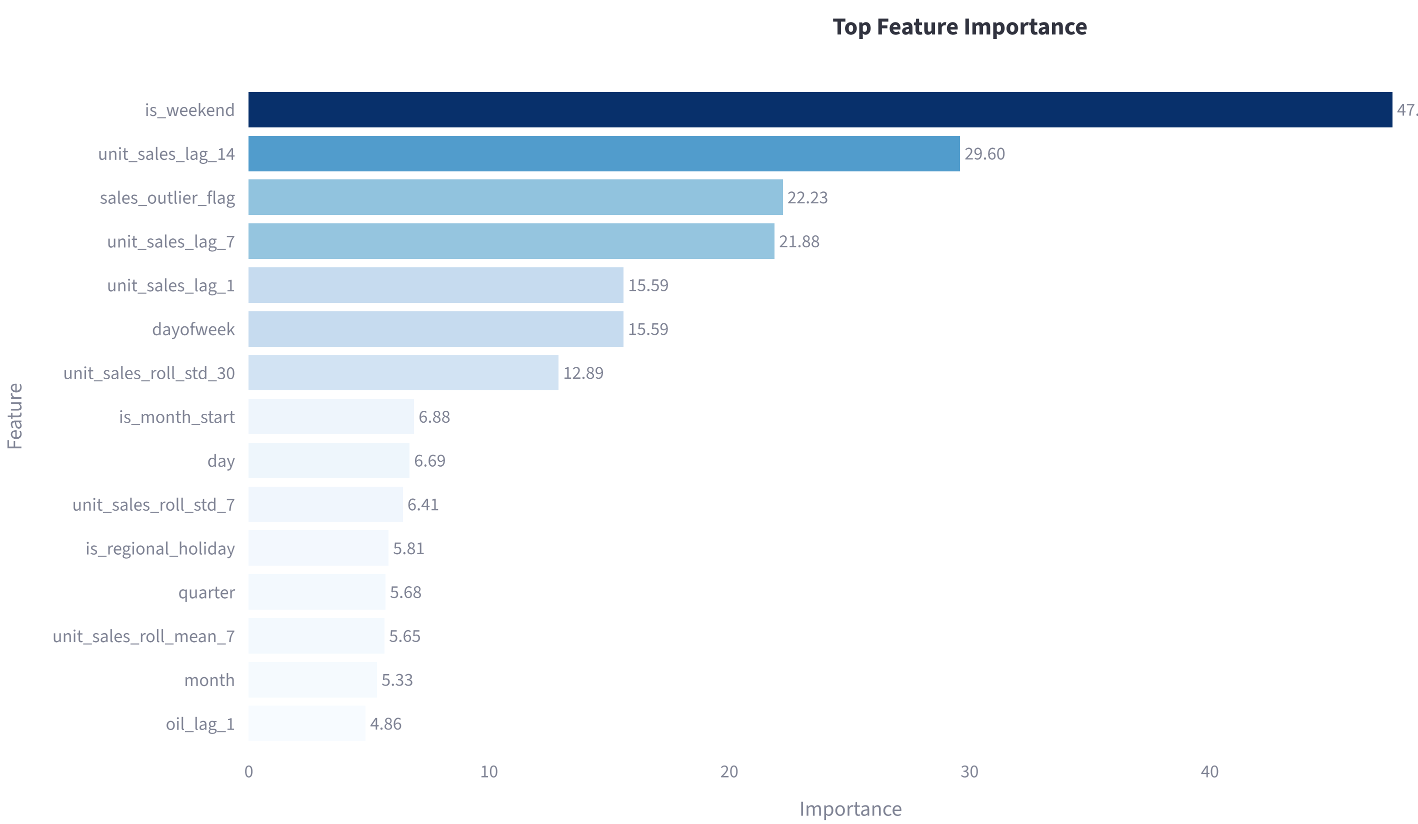
Lower RMSE means better forecasting performance.

Actual vs Predicted Sales



Download Predictions CSV

Feature Importance



Key Business Insights

• Weekend sales consistently exceed weekday sales.

• Lag-based features are strong predictors, confirming that historical sales patterns strongly influence future demand.

• Tuned Ridge Regression achieved the best overall forecasting performance.

• Seasonal and calendar-related effects significantly impact retail sales behavior.

• Forecasting can support better inventory planning, demand preparation, and business decision-making.

Retail Sales Forecasting Dashboard | Built with Streamlit, Plotly, HyperOpt, MLflow, and Scikit-learn